Lab 5: Airline Passenger Satisfaction Prediction with a Neural Network

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The recommended topic was employee attrition. The task allows any other classification problem where a neural network can be used, so I chose **airline passenger satisfaction**: predict from the flight details and service ratings whether a passenger is satisfied or not. The dataset is large (about 130 000 passengers), which is what a neural network needs; on the small employee attrition dataset (1 470 rows) a neural network gives no gain over a simple model.

In [1]:
```
import warnings; warnings.filterwarnings("ignore")
import os; os.environ["TF_CPP_MIN_LOG_LEVEL"] = "3"
import numpy as np, pandas as pd, matplotlib.pyplot as plt, seaborn as sns, time
import tensorflow as tf
from tensorflow import keras
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score,
f1_score, confusion_matrix, ConfusionMatrixDisplay
sns.set_theme(style="whitegrid")
tf.random.set_seed(42); np.random.seed(42);
tf.config.experimental.enable_op_determinism()
print("TensorFlow", tf.__version__, "| Keras", keras.__version__)
```

```
TensorFlow 2.21.0 | Keras 3.15.1
```

1. Dataset Selection

Dataset: Airline Passenger Satisfaction (Klein, 2020), published on Kaggle, license "Other (specified in description)" – free for educational use. **Link:** https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Why it is relevant to my major: airlines collect surveys from every flight and have millions of records. Big Data Analytics turns them into models that tell which service factors really drive satisfaction and which passengers are at risk of leaving to a competitor – the same "customer churn" logic as in Lab 2.

- **Target:** `satisfaction` – "satisfied" (1) or "neutral or dissatisfied" (0). Binary classification.
- **Input features (22):** passenger profile (gender, loyal / disloyal customer, age, type of travel, class), flight (distance, departure and arrival delay in minutes) and 14 service

ratings on a 0–5 scale (wifi, online booking, seat comfort, food, entertainment, leg room, baggage, check-in, cleanliness, …).

- **Observations:** 129 880 passengers (the author gives `train.csv` with 103 904 rows and `test.csv` with 25 976 rows; I join them and make my own split), 25 columns = 22 features + target + 2 index columns.

In [2]:
```
import kagglehub
path = kagglehub.dataset_download("teejmahal20/airline-passenger-satisfaction")
df = pd.concat([pd.read_csv(path + "/train.csv"), pd.read_csv(path +
"/test.csv")], ignore_index=True)
print(df.shape)
df.head()
```

```
(129880, 25)
```

Out[2]:

	Unnamed: 0	id	Gender	Customer Type	Age	Type of Travel	Class	Flight Distance	Inflight wifi service	Depa tim
0	0	70172	Male	Loyal Customer	13	Personal Travel	Eco Plus	460	3	
1	1	5047	Male	disloyal Customer	25	Business travel	Business	235	3	
2	2	110028	Female	Loyal Customer	26	Business travel	Business	1142	2	
3	3	24026	Female	Loyal Customer	25	Business travel	Business	562	2	
4	4	119299	Male	Loyal Customer	61	Business travel	Business	214	3	

5 rows × 25 columns

2. Business Understanding

- **Problem.** An airline knows the satisfaction of a passenger only if the passenger fills in a survey (a few percent of all flights). For the other passengers the airline does not know who is unhappy until they stop flying with it.
- **What the model predicts.** For every passenger and flight – whether the passenger is satisfied, using the ticket data (class, distance, delay) and the ratings that are available.
- **Who will use it.** The customer experience team (which service to fix first), marketing and loyalty programme (whom to send an apology or a bonus after a bad flight), operations (how much a delay really costs in satisfaction).
- **How it helps.** The model shows which factors move satisfaction the most (for example online boarding or wifi), so the airline can spend the budget on the right service.

Predicting unhappy passengers lets the airline react before they leave – a retained frequent flyer is worth thousands of dollars per year.

Success criteria (defined before modelling), on the test set:

Metric	Threshold	Reason
Accuracy	≥ 0.90	must be clearly better than the naive model "everybody is dissatisfied" (57 %) and than a simple linear model (≈ 0.89, see Section 8)
Recall (satisfied class)	≥ 0.90	we must recognise most of the satisfied passengers correctly, otherwise the airline will send apologies to happy clients
F1-score	≥ 0.90	precision and recall must both be high

The most important metric is **F1** (the classes are almost balanced, 43 % vs 57 %, and both types of errors cost money).

Short glossary (definitions follow Géron, 2019, chs. 3 and 10). *Neural network* – a model built from layers of simple units (neurons); each neuron computes a weighted sum of its inputs and passes it through an activation function. *Dense layer* – a layer where every neuron is connected to every input. *ReLU* – activation max(0, x) for hidden layers. *Sigmoid* – activation that squeezes any number into (0, 1), used as the output for binary classification. *Epoch* – one pass through the whole training set. *Batch size* – number of rows the network sees before one weight update. *Learning rate* – the size of one weight update step. *Loss (cost) function* – a number that measures how wrong the predictions are; for binary classification it is *binary cross-entropy*. *Accuracy, precision, recall, F1, confusion matrix* – see Lab 2 (Géron, 2019, ch. 3). All preprocessing is done with scikit-learn (Pedregosa et al., 2011) and the network with Keras / TensorFlow (Chollet, 2015; Abadi et al., 2016).

3. Exploratory Data Analysis

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 25 columns):
 #   Column                            Non-Null Count   Dtype  
---  ------                            --------------   -----  
 0   Unnamed: 0                        129880 non-null  int64  
 1   id                                129880 non-null  int64  
 2   Gender                            129880 non-null  object 
 3   Customer Type                     129880 non-null  object 
 4   Age                               129880 non-null  int64  
 5   Type of Travel                    129880 non-null  object 
 6   Class                             129880 non-null  object 
 7   Flight Distance                   129880 non-null  int64  
 8   Inflight wifi service             129880 non-null  int64  
 9   Departure/Arrival time convenient  129880 non-null  int64  
 10  Ease of Online booking            129880 non-null  int64  
 11  Gate location                     129880 non-null  int64  
 12  Food and drink                    129880 non-null  int64  
 13  Online boarding                   129880 non-null  int64  
 14  Seat comfort                      129880 non-null  int64  
 15  Inflight entertainment            129880 non-null  int64  
 16  On-board service                  129880 non-null  int64  
 17  Leg room service                  129880 non-null  int64  
 18  Baggage handling                  129880 non-null  int64  
 19  Checkin service                   129880 non-null  int64  
 20  Inflight service                  129880 non-null  int64  
 21  Cleanliness                       129880 non-null  int64  
 22  Departure Delay in Minutes        129880 non-null  int64  
 23  Arrival Delay in Minutes          129487 non-null  float64
 24  satisfaction                      129880 non-null  object 
dtypes: float64(1), int64(19), object(5)
memory usage: 24.8+ MB
```

In [4]:
```
df.describe().T.round(2)
```

Out[4]:

	count	mean	std	min	25%	50%	75%	max
Unnamed: 0	129880.0	44158.70	31207.38	0.0	16234.75	38963.5	71433.25	103903.0
id	129880.0	64940.50	37493.27	1.0	32470.75	64940.5	97410.25	129880.0
Age	129880.0	39.43	15.12	7.0	27.00	40.0	51.00	85.0
Flight Distance	129880.0	1190.32	997.45	31.0	414.00	844.0	1744.00	4983.0
Inflight wifi service	129880.0	2.73	1.33	0.0	2.00	3.0	4.00	5.0
Departure/Arrival time convenient	129880.0	3.06	1.53	0.0	2.00	3.0	4.00	5.0
Ease of Online booking	129880.0	2.76	1.40	0.0	2.00	3.0	4.00	5.0
Gate location	129880.0	2.98	1.28	0.0	2.00	3.0	4.00	5.0
Food and drink	129880.0	3.20	1.33	0.0	2.00	3.0	4.00	5.0
Online boarding	129880.0	3.25	1.35	0.0	2.00	3.0	4.00	5.0
Seat comfort	129880.0	3.44	1.32	0.0	2.00	4.0	5.00	5.0
Inflight entertainment	129880.0	3.36	1.33	0.0	2.00	4.0	4.00	5.0
On-board service	129880.0	3.38	1.29	0.0	2.00	4.0	4.00	5.0
Leg room service	129880.0	3.35	1.32	0.0	2.00	4.0	4.00	5.0
Baggage handling	129880.0	3.63	1.18	1.0	3.00	4.0	5.00	5.0
Checkin service	129880.0	3.31	1.27	0.0	3.00	3.0	4.00	5.0
Inflight service	129880.0	3.64	1.18	0.0	3.00	4.0	5.00	5.0
Cleanliness	129880.0	3.29	1.31	0.0	2.00	3.0	4.00	5.0
Departure Delay in Minutes	129880.0	14.71	38.07	0.0	0.00	0.0	12.00	1592.0
Arrival Delay in Minutes	129487.0	15.09	38.47	0.0	0.00	0.0	13.00	1584.0



129 880 rows, 25 columns. Four text columns (gender, customer type, type of travel, class) and the target; all other columns are numeric. `Unnamed: 0` and `id` are index columns without information. `Arrival Delay in Minutes` has 393 missing values.

Some facts: passengers are 7–85 years old (median 40), flights are 31–4 983 miles (median 844), the median delay is 0 minutes but the maximum is 1 592 (26 hours). The 14 service ratings are on a 0–5 scale with means between 2.7 and 3.6; **0 means "not applicable"** (for example wifi rating on a plane without wifi), not the worst grade.

In [5]:
```
print(df["satisfaction"].value_counts(normalize=True).round(3))
fig, ax = plt.subplots(1, 3, figsize=(14, 3.5), sharey=True)
for a, c in zip(ax, ["Class", "Type of Travel", "Customer Type"]):
    df.groupby(c)["satisfaction"].apply(lambda s: (s ==
"satisfied").mean()).plot.bar(ax=a, color="#4C72B0", rot=0, title=f"Share
satisfied by {c}")
plt.tight_layout(); plt.show()
```

```
satisfaction
neutral or dissatisfied    0.566
satisfied                  0.434
Name: proportion, dtype: float64
```



Figure 1. 43 % of passengers are satisfied, 57 % neutral or dissatisfied – the classes are almost balanced. Business class passengers are satisfied in 69 % of cases, economy only in 19 %. Business travellers (58 %) are much happier than personal travellers (10 %), and loyal customers (48 %) happier than disloyal ones (24 %).

In [6]:
```
services = ["Inflight wifi service", "Departure/Arrival time convenient", "Ease of
Online booking", "Gate location", "Food and drink",
            "Online boarding", "Seat comfort", "Inflight entertainment", "On-board
service", "Leg room service", "Baggage handling",
            "Checkin service", "Inflight service", "Cleanliness"]
df["y"] = (df["satisfaction"] == "satisfied").astype(int)
corr = df[services + ["Age", "Flight Distance", "Departure Delay in Minutes",
"Arrival Delay in Minutes", "y"]].corr()["y"].drop("y").sort_values()
plt.figure(figsize=(7, 5))
corr.plot.barh(color="#4C72B0"); plt.title("Correlation of each feature with
satisfaction"); plt.tight_layout(); plt.show()
```

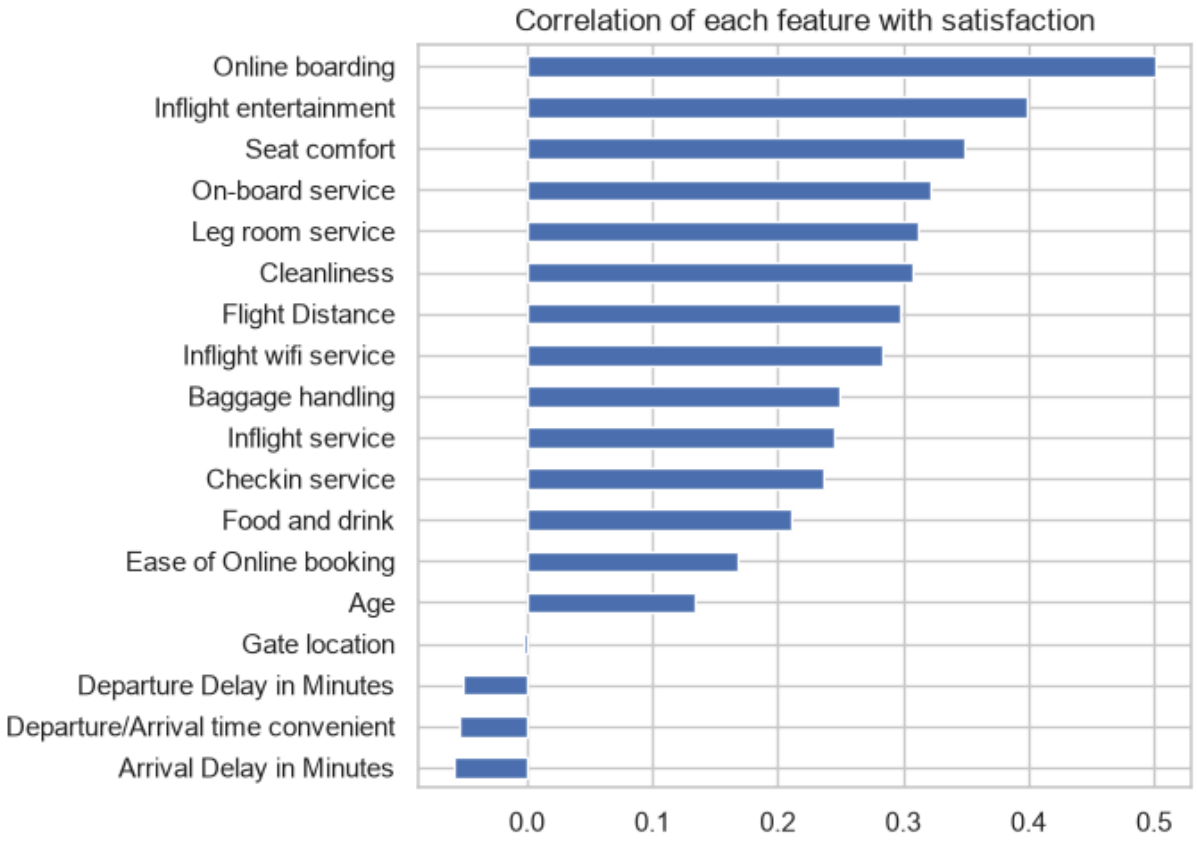


Figure 2. The strongest single factors are online boarding (0.50), inflight entertainment (0.40), seat comfort (0.35) and on-board service (0.32). Delays have almost no linear correlation (−0.05) and gate location none at all. No single feature is decisive – satisfaction is a combination of many ratings, which is a good case for a neural network.

In [7]:
```
fig, ax = plt.subplots(1, 2, figsize=(11, 3.6))
sns.boxplot(data=df, x="satisfaction", y="Online boarding", color="#4C72B0",
ax=ax[0]).set(title="Online boarding rating", xlabel="")
df.groupby(pd.cut(df["Departure Delay in Minutes"], [-1, 0, 15, 60, 180, 2000]))
["y"].mean().plot.bar(ax=ax[1], color="#4C72B0", rot=0, title="Share satisfied by
departure delay (min)")
plt.tight_layout(); plt.show()
```

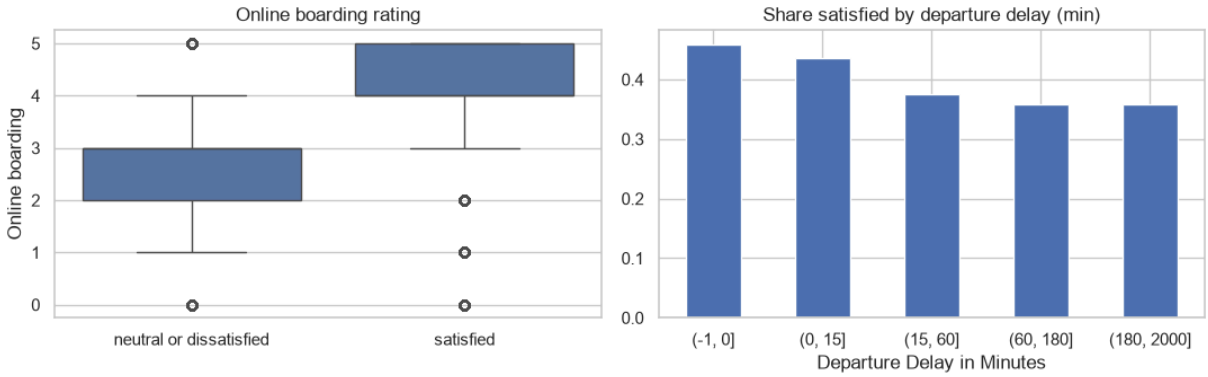


Figure 3. Satisfied passengers rate online boarding 4–5, dissatisfied ones 2–3 – the clearest single split in the data. Delay matters, but not linearly: the share of satisfied passengers is 45

% without delay and drops to 34 % only for delays over 3 hours; short delays (up to 15 minutes) change nothing.

4. Data Cleaning

In [8]:
```
print("missing:", df.isna().sum()[df.isna().sum() > 0].to_dict())
print("duplicate rows (without id):", df.drop(columns=["Unnamed: 0",
"id"]).duplicated().sum(), "| duplicate ids:", df["id"].duplicated().sum())
print("ratings outside 0-5:", int((df[services] > 5).sum().sum() + (df[services] <
0).sum().sum()))
print("age outside 0-100:", (~df["Age"].between(0, 100)).sum(), "| negative
delays:", ((df["Departure Delay in Minutes"] < 0) | (df["Arrival Delay in
Minutes"] < 0)).sum())
print("delay > 6 hours:", (df["Departure Delay in Minutes"] > 360).sum())
print("zero ('not applicable') ratings per column:", {c: int((df[c] == 0).sum())
for c in services if (df[c] == 0).sum() > 100})
```

```
missing: {'Arrival Delay in Minutes': 393}
duplicate rows (without id): 0 | duplicate ids: 0
ratings outside 0-5: 0
age outside 0-100: 0 | negative delays: 0
delay > 6 hours: 150
zero ('not applicable') ratings per column: {'Inflight wifi service': 3916,
'Departure/Arrival time convenient': 6681, 'Ease of Online booking': 5682, 'Food and
drink': 132, 'Online boarding': 3080, 'Leg room service': 598}
```

In [9]:
```
before = df.shape
df = df.drop(columns=["Unnamed: 0", "id", "satisfaction"])
df["Arrival Delay in Minutes"] = df["Arrival Delay in
Minutes"].fillna(df["Departure Delay in Minutes"])
print("shape before:", before, "-> after:", df.shape, "| missing now:",
df.isna().sum().sum())
```

```
shape before: (129880, 26) -> after: (129880, 23) | missing now: 0
```

What was done (129 880 × 25 → 129 880 × 23).

- **Missing values:** 393 arrival delays are missing. Arrival delay is almost the same as departure delay (correlation 0.96), so the missing values are filled with the departure delay of the same flight – better than a global median.
- **Duplicates:** none (no duplicate rows and no duplicate passenger ids).
- **Data types:** correct; the target text column is replaced by the numeric `y`; index columns `Unnamed: 0` and `id` are dropped.
- **Invalid values:** none – all ratings are inside 0–5, ages inside 7–85, no negative delays.
- **Outliers:** 150 flights with a delay over 6 hours are real events (cancellations and re-bookings), they are kept; the scaler and the network handle them. Zero ratings mean "not applicable" (3 000–6 700 zeros for wifi, online booking and time convenience) and are kept as they are; a separate feature counts them (Section 5).

5. Feature Engineering

Created features

Feature	Formula	How it helps the network
`avg_service`	mean of the 14 ratings, zeros ("not applicable") ignored	one compact "overall service" number; correlation with the target 0.53 – higher than any single rating; gives the network a strong direct signal so it can spend its capacity on the interactions
`total_delay`	departure + arrival delay	one delay number instead of two almost identical ones
`n_zero_ratings`	number of ratings equal to 0	tells the network that a 0 is "not applicable", not "terrible"; without it the network would read a zero as the worst grade

Kept: all 14 ratings (the network can learn which ones matter), age, flight distance, the four categorical columns.

Removed: `Unnamed: 0` and `id` (index columns, no information; a network could only memorise them) – already dropped in cleaning. Nothing else is removed: with 130 000 rows and 25 inputs the network is not in danger of over-fitting because of too many features.

In [10]:
```
df["avg_service"] = df[services].replace(0, np.nan).mean(axis=1).fillna(0)
df["total_delay"] = df["Departure Delay in Minutes"] + df["Arrival Delay in
Minutes"]
df["n_zero_ratings"] = (df[services] == 0).sum(axis=1)
print(df[["avg_service", "total_delay", "n_zero_ratings", "y"]].corr()
["y"].round(2))
print(df.shape)
```

```
avg_service       0.53
total_delay      -0.05
n_zero_ratings    0.11
y                 1.00
Name: y, dtype: float64
(129880, 26)
```

6. Data Preprocessing

- X = 25 features, y = `y`. First split 80 / 20 into train and **test** (stratified, `random_state=42`); then 20 % of the training part is kept as a **validation set** to watch the loss during training and to stop early. The test set is used only once, in Section 8.
- Categories → OneHotEncoder (binary columns like gender become one 0/1 column). Numeric features → StandardScaler. **Scaling is essential for a neural network:** gradient descent works badly when one input is in thousands (flight distance) and another in 0–5 (ratings); with unscaled inputs the training is slow and often does not converge.
- Encoder and scaler are fitted on the training part only – no data leakage. Keras needs plain NumPy arrays, so the transformed data is stored as arrays.

In [11]:
```
X = df.drop(columns=["y"])
y = df["y"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42, stratify=y)
X_train, X_val, y_train, y_val = train_test_split(X_train, y_train, test_size=0.2,
random_state=42, stratify=y_train)
cat_cols = ["Gender", "Customer Type", "Type of Travel", "Class"]
num_cols = [c for c in X.columns if c not in cat_cols]
prep = ColumnTransformer([("num", StandardScaler(), num_cols), ("cat",
OneHotEncoder(drop="if_binary", sparse_output=False), cat_cols)]).fit(X_train)
A_train, A_val, A_test = prep.transform(X_train), prep.transform(X_val),
prep.transform(X_test)
n_inputs = A_train.shape[1]
print("train:", A_train.shape, "| validation:", A_val.shape, "| test:",
A_test.shape)
print("satisfied share train / val / test: %.3f / %.3f / %.3f" % (y_train.mean(),
y_val.mean(), y_test.mean()))
```

```
train: (83123, 27) | validation: (20781, 27) | test: (25976, 27)
satisfied share train / val / test: 0.434 / 0.434 / 0.434
```

7. Neural Network Architecture and Training

How the network works.

- **Forward propagation.** The 27 scaled inputs enter the first layer. Every neuron of a hidden layer computes $z = w \cdot x + b$ (weighted sum of its inputs plus a bias) and applies ReLU: $a = max(0, z)$. The outputs of one layer are the inputs of the next. The last neuron applies the sigmoid, $\sigma(z) = 1 / (1 + e^{-z})$, which turns the number into a probability of "satisfied". If the probability is above 0.5 the prediction is "satisfied".
- **Loss function.** *Binary cross-entropy* compares the predicted probability p with the true label y: $loss = -[y \cdot log(p) + (1 - y) \cdot log(1 - p)]$. It is close to 0 when the probability is right and grows fast when the network is confidently wrong. The loss is averaged over all rows in a batch.
- **Backpropagation.** After the forward pass the network computes the gradient of the loss with respect to every weight – how much the loss would change if the weight moved a little. It does this layer by layer from the output back to the input, using the chain rule (Rumelhart et al., 1986; Géron, 2019, ch. 10).
- **Gradient descent.** Every weight is moved a small step against its gradient: $w \leftarrow w - \eta \cdot \partial loss / \partial w$, where η is the **learning rate**. Here the **Adam** optimizer (Kingma & Ba, 2015) is used – gradient descent with an adaptive step for every weight. One update per batch; one epoch = all batches.

Baseline architecture (three layers).

Layer	Neurons	Activation
Input	27 (the preprocessed features)	–
Hidden 1 (Dense)	32	ReLU
Output (Dense)	1	Sigmoid

Trained with Adam, learning rate 0.001, batch size 128, up to 30 epochs with **early stopping** (training stops when the validation loss does not improve for 5 epochs and the best weights are restored).

In [12]:
```
def build(hidden=(32,), lr=0.001):
    layers = [keras.Input(shape=(n_inputs,))] + [keras.layers.Dense(h,
activation="relu") for h in hidden] + [keras.layers.Dense(1,
activation="sigmoid")]
    model = keras.Sequential(layers)
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=lr),
loss="binary_crossentropy", metrics=["accuracy"])
    return model

def train(model, epochs=30, batch_size=128):
    es = keras.callbacks.EarlyStopping(patience=5, restore_best_weights=True)
    t = time.time()
    h = model.fit(A_train, y_train, validation_data=(A_val, y_val), epochs=epochs,
batch_size=batch_size, verbose=0, callbacks=[es])
    return h, time.time() - t

def test_metrics(model):
    p = (model.predict(A_test, verbose=0) > 0.5).astype(int).ravel()
    return {"Accuracy": accuracy_score(y_test, p), "Precision":
precision_score(y_test, p), "Recall": recall_score(y_test, p), "F1":
f1_score(y_test, p)}

baseline = build((32,))
baseline.summary()
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please
use WSL2 or the TensorFlow-DirectML plugin.
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 32)	896
dense_1 (Dense)	(None, 1)	33

Total params: 929 (3.63 KB)

Trainable params: 929 (3.63 KB)

Non-trainable params: 0 (0.00 B)

In [13]:
```
loss0, acc0 = baseline.evaluate(A_train, y_train, verbose=0)
w0 = baseline.layers[0].get_weights()[0].copy()
print("BEFORE training: loss = %.3f, accuracy = %.3f (random weights)" % (loss0,
acc0))
hist, t = train(baseline)
loss1, acc1 = baseline.evaluate(A_train, y_train, verbose=0)
w1 = baseline.layers[0].get_weights()[0]
print("AFTER training:  loss = %.3f, accuracy = %.3f | epochs: %d | time: %.0f s"
% (loss1, acc1, len(hist.history["loss"]), t))
print("weights of hidden layer 1: shape %s | mean |change| = %.3f | std before
%.3f -> after %.3f" % (w0.shape, np.abs(w1 - w0).mean(), w0.std(), w1.std()))
res = {"Baseline NN (32)": test_metrics(baseline)}
print("test:", {k: round(v, 4) for k, v in res["Baseline NN (32)"].items()})
```

```
BEFORE training: loss = 0.744, accuracy = 0.539 (random weights)
AFTER training:  loss = 0.104, accuracy = 0.957 | epochs: 30 | time: 26 s
weights of hidden layer 1: shape (27, 32) | mean |change| = 0.248 | std before 0.181
-> after 0.434
test: {'Accuracy': 0.9512, 'Precision': 0.9553, 'Recall': 0.9312, 'F1': 0.9431}
```

Loss before and after training. With random initial weights the loss is about 0.74 and accuracy about 0.54 – close to a coin flip, the network only guesses. After training the loss on the training set falls to about 0.10 and accuracy rises to about 0.96. The 27 × 32 weights of the first hidden layer changed on average by about 0.25 (their spread grew from 0.18 to about 0.43): gradient descent moved every weight many times, each step in the direction that reduced the loss.

In [14]:
```
fig, ax = plt.subplots(1, 2, figsize=(11, 3.5))
ax[0].plot(hist.history["loss"], label="train");
ax[0].plot(hist.history["val_loss"], label="validation");
ax[0].set(title="Baseline: loss per epoch", xlabel="epoch", ylabel="binary cross-
entropy"); ax[0].legend()
ax[1].plot(hist.history["accuracy"], label="train");
ax[1].plot(hist.history["val_accuracy"], label="validation");
ax[1].set(title="Baseline: accuracy per epoch", xlabel="epoch"); ax[1].legend()
plt.tight_layout(); plt.show()
```

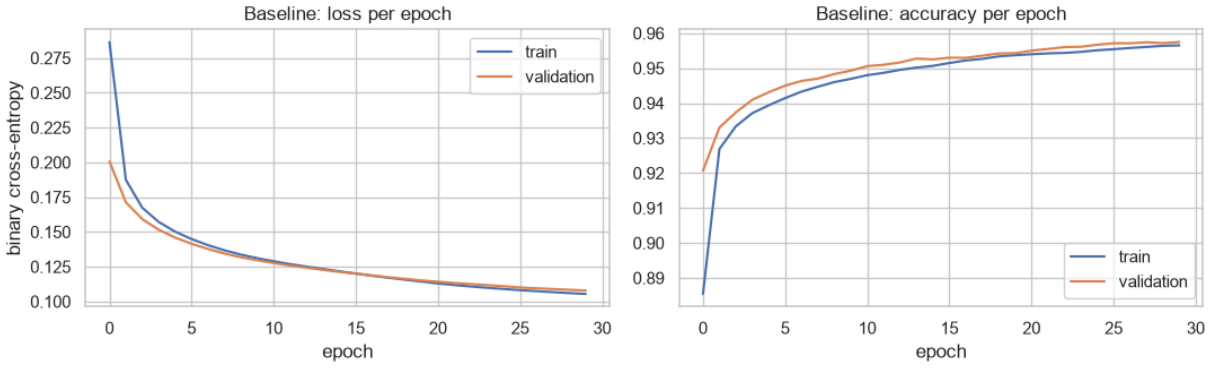


Figure 4. Both losses fall quickly in the first 5 epochs and then slowly; the validation curve stays very close to the training curve, so the baseline does not over-fit. The curves are still going down at epoch 30 – a small network with one hidden layer learns slowly, which suggests trying more neurons and layers.

7.1 Hyperparameter experiments

Each experiment changes one thing compared to the configuration (64, 32) neurons, learning rate 0.001, batch 128, up to 30 epochs. Every model is scored on the test set.

In [15]:
```
def experiment(name, hidden=(64, 32), lr=0.001, batch_size=128, epochs=30):
    m = build(hidden, lr); h, t = train(m, epochs, batch_size)
    r = test_metrics(m); r.update({"epochs run": len(h.history["loss"]), "best val loss": min(h.history["val_loss"]), "time, s": t})
    return name, r, m, h

exp = []
for hidden in [(32,), (64, 32), (128, 64, 32)]:
    exp.append(experiment(f"hidden layers {hidden}", hidden=hidden))
pd.DataFrame({n: r for n, r, _, _ in exp}).T.round(4)
```

Out[15]:

	Accuracy	Precision	Recall	F1	epochs run	best val loss	time, s
hidden layers (32,)	0.9511	0.9583	0.9278	0.9428	30.0	0.1105	24.4855
hidden layers (64, 32)	0.9574	0.9665	0.9343	0.9501	30.0	0.0966	26.6270
hidden layers (128, 64, 32)	0.9572	0.9752	0.9251	0.9495	13.0	0.1002	13.6658

Number of hidden layers and neurons. One hidden layer of 32 neurons gives F1 ≈ 0.94; two layers (64, 32) improve it to ≈ 0.95 and reach a lower validation loss (≈ 0.097 vs 0.111). A third layer (128, 64, 32) gives the same quality (F1 ≈ 0.95) and converges in fewer epochs (13), but its validation loss is not better – the extra capacity is not needed for 27 inputs. **(64, 32) is the best size.**

In [16]:
```
exp = []
for lr in [0.0001, 0.001, 0.01]:
    exp.append(experiment(f"learning rate {lr}", lr=lr))
pd.DataFrame({n: r for n, r, _, _ in exp}).T.round(4)
```

Out[16]:

	Accuracy	Precision	Recall	F1	epochs run	best val loss	time, s
learning rate 0.0001	0.9457	0.9512	0.9224	0.9366	30.0	0.1220	28.2832
learning rate 0.001	0.9568	0.9542	0.9460	0.9500	30.0	0.0961	27.5235
learning rate 0.01	0.9580	0.9692	0.9329	0.9507	17.0	0.0917	15.5715

Learning rate. With 0.0001 the steps are too small: after 30 epochs the loss is still about 0.122 and the model is the weakest (under-training). 0.001 (the Adam default) and 0.01 reach the same F1 ≈ 0.95; 0.01 converges in fewer epochs, but a large rate can jump over the minimum on other data, so 0.001 stays the safe choice.

In [17]:
```
exp = []
for bs in [32, 128, 512]:
    exp.append(experiment(f"batch size {bs}", batch_size=bs))
pd.DataFrame({n: r for n, r, _, _ in exp}).T.round(4)
```

Out[17]:

	Accuracy	Precision	Recall	F1	epochs run	best val loss	time, s
batch size 32	0.9576	0.9735	0.9276	0.9500	25.0	0.0971	80.3328
batch size 128	0.9584	0.9649	0.9383	0.9514	27.0	0.0950	24.3710
batch size 512	0.9557	0.9677	0.9289	0.9479	30.0	0.1000	10.8175

Batch size. Batch 32 makes ≈ 2 600 weight updates per epoch – the most accurate gradient path, but the slowest (about 7 times longer than batch 512: 80 s vs 11 s). Batch 512 is the fastest but the loss is a little higher because fewer updates are made. Batch 128 is a good compromise: almost the same F1 as 32 at a fraction of the time.

In [18]:
```
exp = []
for ep in [5, 30, 100]:
    exp.append(experiment(f"max epochs {ep}", hidden=(64, 32), epochs=ep))
final_name, final_res, final_model, final_hist = exp[-1]
pd.DataFrame({n: r for n, r, _, _ in exp}).T.round(4)
```

Out[18]:

	Accuracy	Precision	Recall	F1	epochs run	best val loss	time, s
max epochs 5	0.9514	0.9554	0.9316	0.9433	5.0	0.1118	5.2531
max epochs 30	0.9587	0.9618	0.9424	0.9520	30.0	0.0958	27.4663
max epochs 100	0.9575	0.9646	0.9365	0.9503	34.0	0.0977	31.2175

Number of epochs. 5 epochs are not enough (loss ≈ 0.112, F1 ≈ 0.94 – under-fitting). With up to 100 epochs early stopping ends training after 34 epochs anyway, because the validation loss stops improving; the training loss keeps going down, so more epochs would

only give over-fitting. Early stopping is therefore the right way to choose the number of epochs.

Final model: 27 inputs → Dense 64 (ReLU) → Dense 32 (ReLU) → Dense 1 (Sigmoid); Adam, learning rate 0.001, batch 128, early stopping (the model from the last experiment).

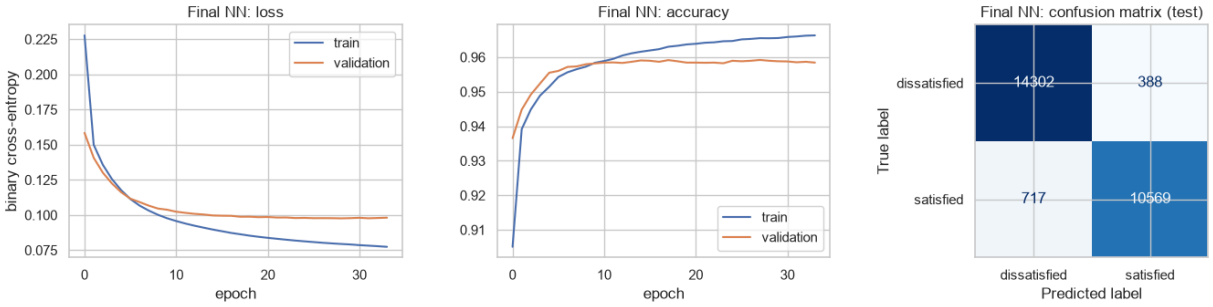
8. Evaluation

In [19]:
```
logreg = LogisticRegression(max_iter=3000).fit(A_train, y_train)
p_lr = logreg.predict(A_test)
res["Logistic Regression (reference)"] = {"Accuracy": accuracy_score(y_test,
p_lr), "Precision": precision_score(y_test, p_lr), "Recall": recall_score(y_test,
p_lr), "F1": f1_score(y_test, p_lr)}
res["Final NN (64, 32)"] = {k: final_res[k] for k in ["Accuracy", "Precision",
"Recall", "F1"]}
table = pd.DataFrame(res).T.round(4)
table["meets criteria"] = (table["Accuracy"] >= 0.90) & (table["Recall"] >= 0.90)
& (table["F1"] >= 0.90)
table
```

Out[19]:

	Accuracy	Precision	Recall	F1	meets criteria
Baseline NN (32)	0.9512	0.9553	0.9312	0.9431	True
Logistic Regression (reference)	0.8871	0.8716	0.8681	0.8698	False
Final NN (64, 32)	0.9575	0.9646	0.9365	0.9503	True

In [20]:
```
fig, ax = plt.subplots(1, 3, figsize=(14, 3.6))
ax[0].plot(final_hist.history["loss"], label="train");
ax[0].plot(final_hist.history["val_loss"], label="validation");
ax[0].set(title="Final NN: loss", xlabel="epoch", ylabel="binary cross-entropy");
ax[0].legend()
ax[1].plot(final_hist.history["accuracy"], label="train");
ax[1].plot(final_hist.history["val_accuracy"], label="validation");
ax[1].set(title="Final NN: accuracy", xlabel="epoch"); ax[1].legend()
p_nn = (final_model.predict(A_test, verbose=0) > 0.5).astype(int).ravel()
ConfusionMatrixDisplay(confusion_matrix(y_test, p_nn), display_labels=
["dissatisfied", "satisfied"]).plot(ax=ax[2], cmap="Blues", colorbar=False);
ax[2].set_title("Final NN: confusion matrix (test)")
plt.tight_layout(); plt.show()
print("epochs trained:", len(final_hist.history["loss"]), "| final train loss %.3f
| best validation loss %.3f" % (final_hist.history["loss"][-1],
min(final_hist.history["val_loss"])))
```



`epochs trained: 34 | final train loss 0.077 | best validation loss 0.098`

Over-fitting or under-fitting? The training loss keeps falling (to ≈ 0.077) while the validation loss flattens at ≈ 0.098 after about 10 epochs and then stays level – a small gap appears, which is the beginning of over-fitting, and early stopping stops the training exactly there (after 34 epochs). Train accuracy (≈ 0.97) and test accuracy (≈ 0.96) are close, so the model is neither under-fitted (as the 5-epoch or lr = 0.0001 versions were) nor seriously over-fitted.

Results. The final network reaches accuracy ≈ 0.958, precision ≈ 0.96, recall ≈ 0.94 and F1 ≈ 0.95 on 25 976 test passengers. Of 11 286 satisfied passengers about 10 600 are recognised, and only about 400 of 14 690 dissatisfied passengers are wrongly called satisfied. The reference logistic regression on the same inputs gives accuracy ≈ 0.887 and F1 ≈ 0.870 – the network adds 7–8 points, because satisfaction depends on **combinations** of ratings and passenger type (for example, online boarding matters for business travellers, wifi for economy), and a linear model cannot learn such interactions.

Business criteria. Accuracy ≈ 0.958 ≥ 0.90, recall ≈ 0.94 ≥ 0.90, F1 ≈ 0.95 ≥ 0.90 – all met. The logistic regression fails all three.

9. Conclusion

- **Data and preprocessing.** 129 880 airline passengers (Kaggle), 22 features, 43 % satisfied. 393 missing arrival delays filled with the departure delay, index columns removed, no duplicates or invalid values. Three features created (`avg_service` , `total_delay` , `n_zero_ratings`), categories one-hot encoded, numbers standardised on the training set only; 80 / 20 train / test split plus a validation set for early stopping.
- **Architecture and training.** Keras sequential network 27 → 64 (ReLU) → 32 (ReLU) → 1 (Sigmoid), binary cross-entropy loss, Adam optimizer, learning rate 0.001, batch 128, early stopping after 34 epochs. Loss went from about 0.74 (random weights) to about 0.08 on the training set; the first-layer weights changed on average by about 0.25.
- **Hyperparameters.** Two hidden layers beat one; a third adds nothing. Learning rate 0.0001 under-trains, 0.001 and 0.01 are equally good. Batch 128 balances accuracy and speed. Early stopping picks the number of epochs; 5 epochs are too few.

- **Results.** Final network: accuracy ≈ 0.958, F1 ≈ 0.95, recall ≈ 0.94 on the test set; logistic regression: accuracy ≈ 0.887, F1 ≈ 0.870. **All business criteria are met.** No serious over-fitting: train and validation curves stay close.
- **Improvements.** (1) Add dropout or L2 regularization and a learning-rate schedule to close the small train / validation gap; (2) analyse the network with SHAP or permutation importance to tell the airline exactly which services to improve for which passenger group; (3) collect real survey data with dates and routes and check the model on new months, because passenger expectations change over time.

10. References

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